# Math 10 - Functions and Relations



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

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Last Updated: January 22, 2026

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# 1 Relations and Functions

## Definitions

A ***set*** is a collection of distinct objects. An ***element*** of a set is one object in the set. A ***relation*** associates the elements of one set with the elements of another set.

Relations are sets of ordered pairs. The set of first components, or $x$-values, is the ***domain*** of the relation. The set of second components, or $y$-values, is called the ***range*** of the relation.

To find solutions to a relation, values are arbitrarily assigned for $x$ from the set of real numbers. This means $x$ is the **independent variable.** Choosing arbitrary "input" values for $x$ produces "output" values for $y$. The output value depends on the value of $x$, therefore it is called the **dependent variable**.

**Example 1.1.** Represent the solutions to the relation below in different forms.

| Input $x$ | Relation $y = 2x + 1$ | Output $y$ |
|---|---|---|
| $-3$ | $2(-3) + 1$ | $-5$ |
| $0$ | $2(0) + 1$ | $1$ |
| $2$ | $2(2) + 1$ | $5$ |

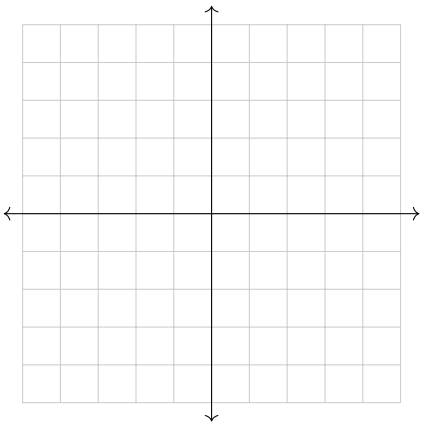
These values represent three solutions to infinitely many solutions of the relation $y = 2x + 1$. The solutions can also be presented in the following ways

**Ordered Pairs**

**Table**

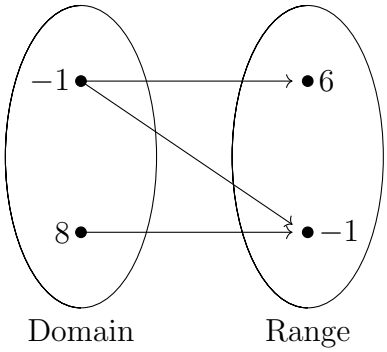
**Mapping Diagram**

**Graph**

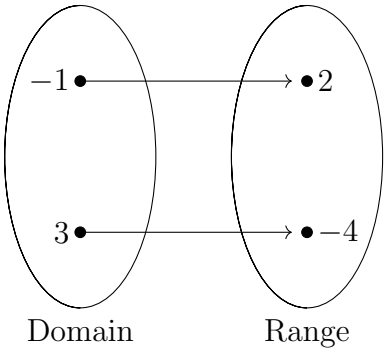


**Example 1.2.** Write the mapping diagram in ordered pair notation.

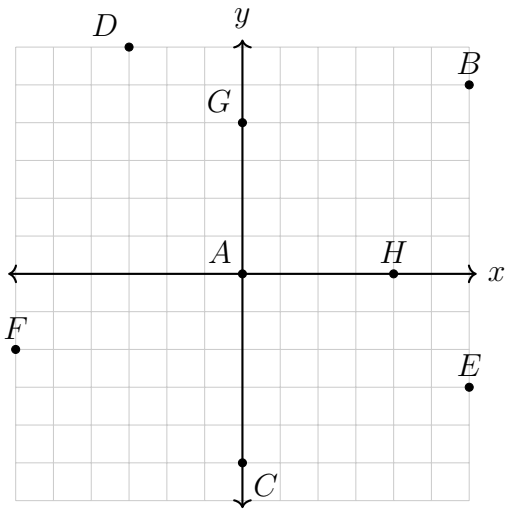
a)




b)




**Example 1.3.** Determine the coordinates of each point and then the domain and the range of each coordinate.



## Functions

A function is a relation for which each input has exactly one output. If any input of a relation has more than one output, the relation is not a function.

"Every function is a relation but not every relation is a function."

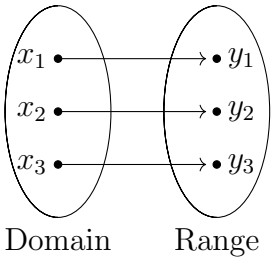
**Determining whether a relation is a function:**

Look at every input and make sure it only has one output. Make sure the input is NOT repeated.

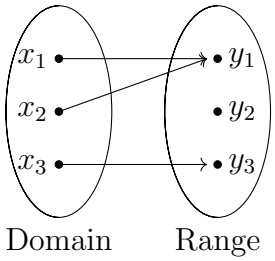
**Note:** An output may be repeated for two or more different inputs.

**Example 1.4.** Determine whether the mapping diagram below represents a function.

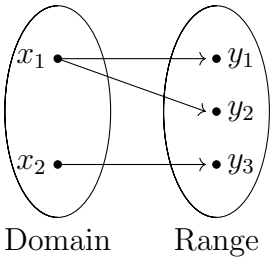
a)




b)




c)

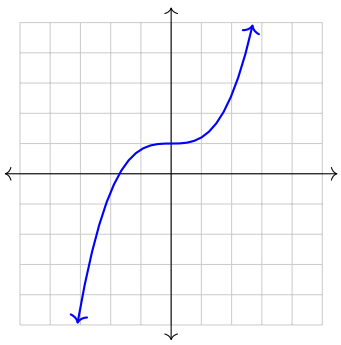



### Vertical Line Test

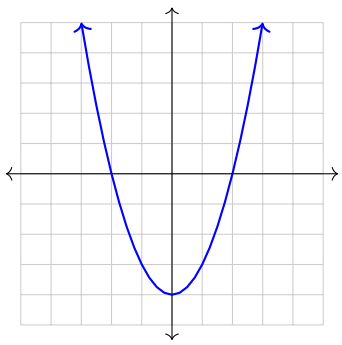
An equation defines $y$ as a function of $x$ if and only if every vertical line in the coordinate plane intersects the graph of the equation at most once.

**Example 1.5.** Determine whether each of the following relations passes the vertical line test.

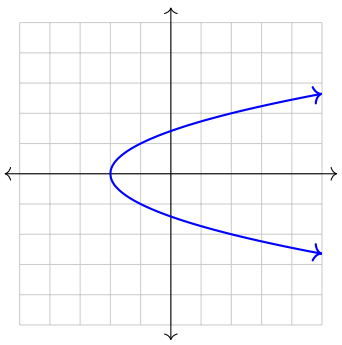
a)



b)



c)

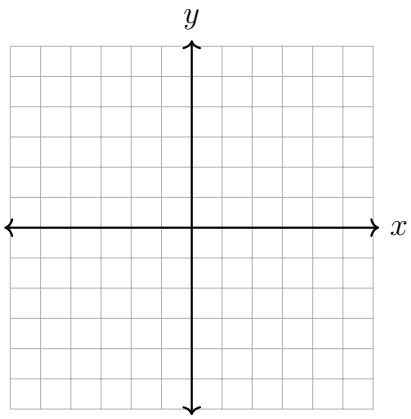


## Practice Exercises

1. Plot and label following points on the grid below

$$A(2,3), \quad B(5,5), \quad C(0,-2), \quad D(0,5)$$

$$E(3,0), \quad F(3,2), \quad G(-3,-3), \quad H(-1,0)$$



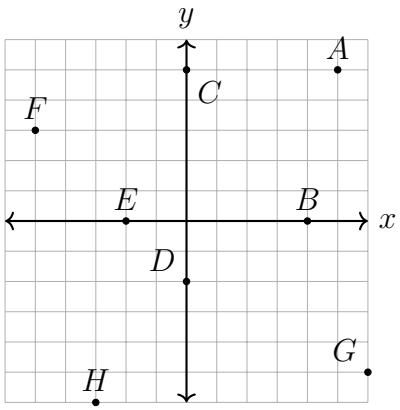

2. Write down the coordinates of all the points

$A($ $,$ $)$ $B($ $,$ $)$ $C($ $,$ $)$

$D($ $,$ $)$ $E($ $,$ $)$

$F($ $,$ $)$ $G($ $,$ $)$

$H($ $,$ $)$




3. (Multiple-Choice) A relation is:

   A. Any set of ordered pairs.

   B. Two sets of ordered pairs that are related.

   C. A graph of ordered pairs.

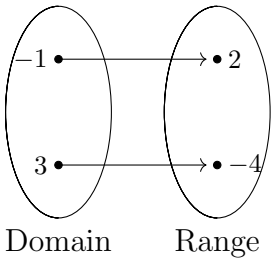
   D. A set of ordered pairs where the domain corresponds to exactly 1 range.

4. (Multiple-Choice) The domain of a relation is:
   A. The sum of all $x$ and $y$ values in ordered pairs.
   B. The sum of the components in the ordered pairs.
   C. The set of all first components in the ordered pairs.
   D. The set of all second components in the ordered pairs.

5. (Multiple-Choice) The range of a relation is:
   A. The sum of all $x$ and $y$ values in ordered pairs.
   B. The sum of the components in the ordered pairs.
   C. The set of all first components in the ordered pairs.
   D. The set of all second components in the ordered pairs.

6. Write the mapping diagram in ordered pair notation.

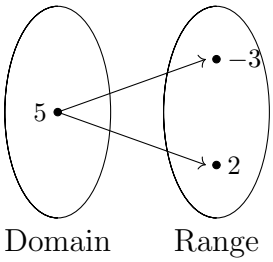
a)




( , )

( , )

b)




( , )

( , )

7. Draw a mapping diagram for the ordered pairs.

a) $(1, 3), (2, 3), (4, 3)$

b) $(1, 3), (2, 2), (3, 1)$

c) $(3,1),(3,2),(3,3)$

d) $(1,3),(1,2),(2,3)$

8. (Multiple-Choice) A function is:
   A. Any set of ordered pairs.
   B. A set of ordered pairs in which every value in the domain corresponds to exactly one value in the range.
   C. A set of ordered pairs in which the range corresponds to exactly one domain.
   D. A graph of ordered pairs.

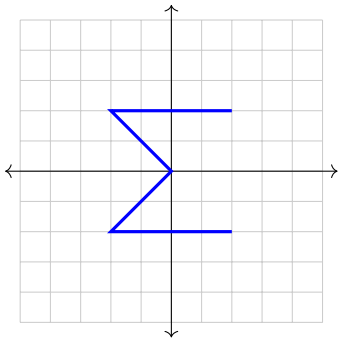
9. (Multiple-Choice) Which sets of ordered pairs represents a function?
   A. $(3,3),(-3,3),(5,-2),(-5,-2)$
   B. $(3,3),(3,-3),(-2,5),(-2,-5)$
   C. $(2,2),(2,3),(2,4)$
   D. $(2,2),(3,2),(4,2)$

10. (Multiple-Choice) Which is a description of a function?
    A. For each domain, there is exactly one range.
    B. For each range, there is exactly one domain.

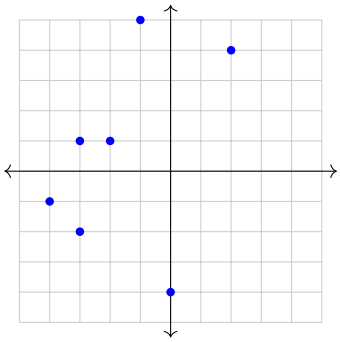
11. (Multiple-Choice) Which is a description of a range?
    A. The set of inputs.
    B. The set of outputs.

12. Do the following graphs pass the vertical line test? Hence, is it a function?

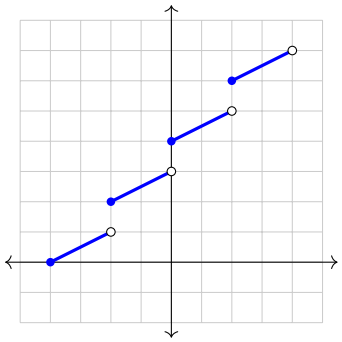
a)



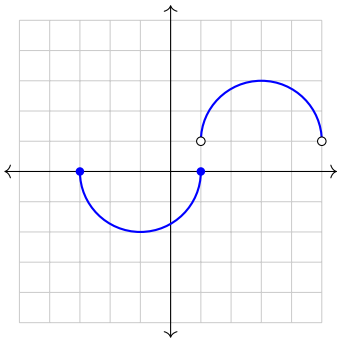
b)



c)

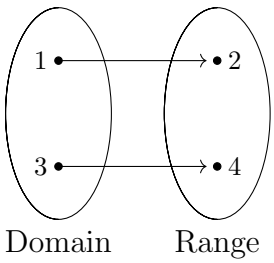


d)

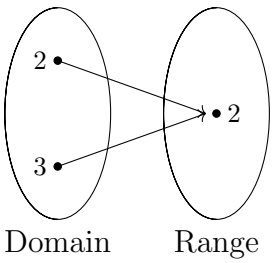


13. Do the following mapping diagrams indicate functions?

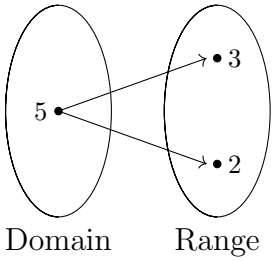
a)



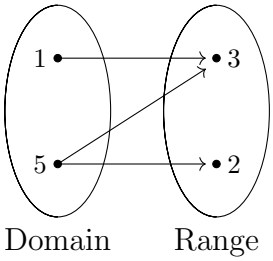

b)




c)




d)



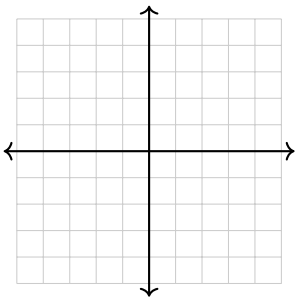
14. Use the relation defined by the table to answer the questions below.

| $x$ | $-4$ | $-2$ | $0$ | $2$ |
|---|---|---|---|---|
| $y$ | $2$ | $0$ | $-2$ | $-4$ |

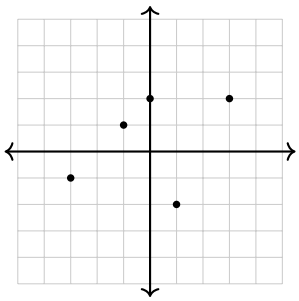
a) Express the relation using a mapping diagram and determine whether it is a function.

b) Express the relation using ordered pair notation.

c) Graph this relation.



15. Use the graph to answer the questions below.



a) Express the relation using a mapping diagram and determine whether it is a function.

b) Express the relation using ordered pair notation.

c) Express this relation using a table format.

| $x$ | | | | | |
|---|---|---|---|---|---|
| $y$ | | | | | |

# 2 Domain and Range Notation

The set of first elements of a relation is called the **domain**. The domain is the set of all possible inputs ($x$-values).

The set of second elements of a relation is called the **range**. The range is the set of all possible outputs ($y$-values).
One way to write a set is to list is elements inside braces.
**Example 2.1.** Determine the domain and the range of the ordered pairs.

$$A = \{(1, 2), (-3, 5), (4, -2)\}$$
$$B = \{(-3, 4), (1, 0), (0, 2), (3, 2)\}$$
$$C = \{(-2, 1), (1, 0), (3, 3), (1, 4)\}$$

| | |
|---|---|
| Domain of A: | Range of A: |
| Domain of B: | Range of B: |
| Domain of C: | Range of C: |

**Note:** In defining a domain or range, we *omit* duplicates.

## Notations

When we want to express domain and range, there are a few different notations that we can choose.

One is the **set notation** that we used above. Sets are denoted by {}, and your elements are inside the set. Set notation is great when you have discrete (not continuous) elements. For example, the set of natural numbers 1 to 8 can be written as

$$\{1, 2, 3, 4, 5, 6, 7, 8\}$$

Another common notation used when you have a range of numbers (continuous), is **inequality notation**. This is something most students are familiar with.

The last notation we will look at is called **interval notation**, and this is great when you have start and end values in your range.

### Inequality Notation

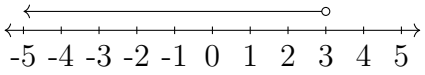
Here are a few different inequality forms that we use.

| | |
|---|---|
| $x < a$ | $x$ is less than $a$ |
| $x > a$ | $x$ is greater than $a$ |
| $a < x < b$ | $x$ greater than $a$ and less than $b$ |

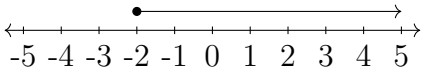
We can also use $\leq$ and $\geq$ to also indicate "or equal to".

**Example 2.2.** Each number line represents the possible values of $x$. Write the inequality represented. An open circle means cannot equal to.

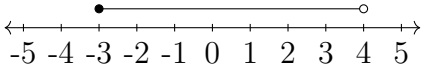
a)



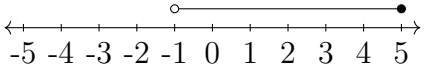

b)




c)

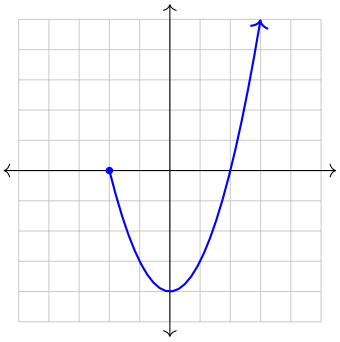



d)




**Example 2.3.** Find the domain and range of the following. Express using **inequality notation**.

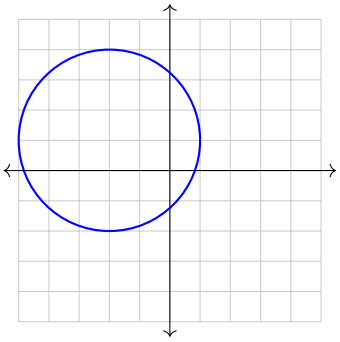
a)



Domain:

Range:

b)



Domain:

Range:

### Interval Notation

Interval notation is great because it's clean and simple to write.

| Inequality Notation | Interval Notation |
|---|---|
| $x < a$ | $(-\infty, a)$ |
| $x \leq a$ | $(-\infty, a]$ |
| $x > a$ | $(a, \infty)$ |
| $x \geq a$ | $[a, \infty)$ |
| $a < x < b$ | $(a, b)$ |
| $a \leq x \leq b$ | $[a, b]$ |
| $a \leq x < b$ | $[a, b)$ |
| $a < x \leq b$ | $(a, b]$ |

It may appear that the interval notation for $a < x < b$ is $(a, b)$, which looks a lot like our coordinate notation for a point $(a, b)$. This confusion is usually cleared based on **context**.

### The Symbol $\in$

Some of you may have seen this symbol before. It is mathematically used to show that something is an element of some set. For example,

$$2 \in (1, 4)$$

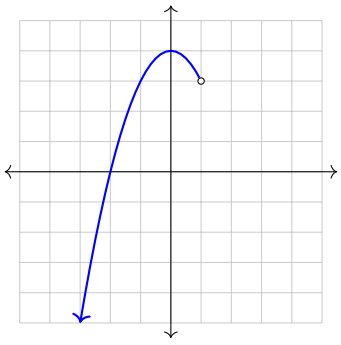
The context here tells me that 2 ***is an element of*** the set of numbers between 1 and 4.

More commonly, we will simply say "2 ***is in*** the interval from 1 to 4".

So if you want to remove ambiguity, then we can use $x \in (a, b)$ and $y \in (c, d)$ to really be clear we're expressing domain and range in interval notation. When explicitly answering domain and range questions, you can write just the interval. Using $x \in$ or $y \in$ is optional.

**Example 2.4.** Find the domain and range of the following. Express using **interval notation**.

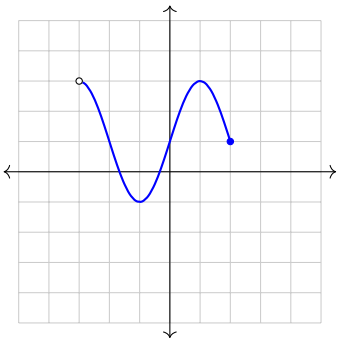
a)



Domain:

Range:

b)



Domain:

Range:

### Set-Builder Notation

This is the last notation that we use and although it's the most complicated, it's also the most informative. It also uses the set notation of {}.

The usual format of set-builder notation is

$$\{A \mid B\}$$

Where $A$ is what variable and type of numbers you're working with, and $B$ is any conditions on that variable.

For example,

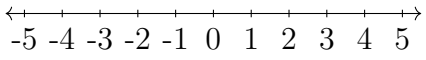
$$\{y \in \mathbb{N} \mid y \geq 10\}$$

the above reads, $y$ is a natural number, ***given that*** $y$ is greater than or equal to 10. This is equivalent to $\{10, 11, 12, \dots\}$, but without the assumption that the pattern continues as implied.

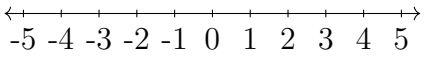
If no number set is specified, we assume it is over the set of all real numbers, $\mathbb{R}$.

**Example 2.5.** Plot the values on the number line indicated by the set notation.

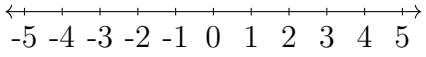
a) $\{x \mid 1 \leq x < 5\}$



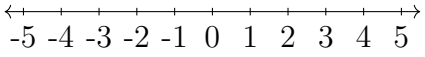

b) $\{x \in \mathbb{Z} \mid x \leq 2\}$




c) $\{x \mid x \neq -1\}$

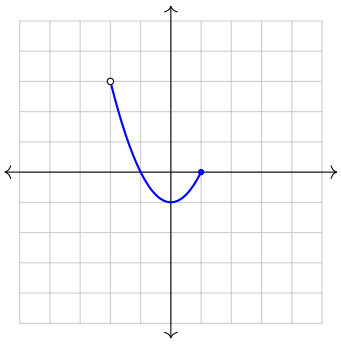



d) $\{x \in \mathbb{N} \mid x < 5\}$




**Example 2.6.** Find the domain and range of the following. Express using **set-builder notation**.

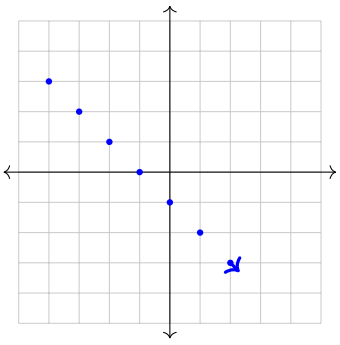
a)



Domain:

Range:

b)



Domain:

Range:

### No Restrictions

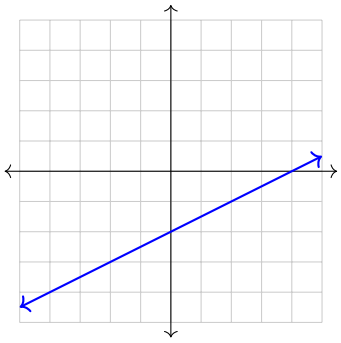
If you ever come across trying to express something with no restrictions, ie. all real numbers. We have a multitude of ways to show this.

$$\mathbb{R}, \quad (-\infty, \infty), \quad x \in \mathbb{R}, \quad y \in \mathbb{R}$$

Are all valid ways to say any real number is fine.

**Example 2.7.** Find the domain and range of the following.

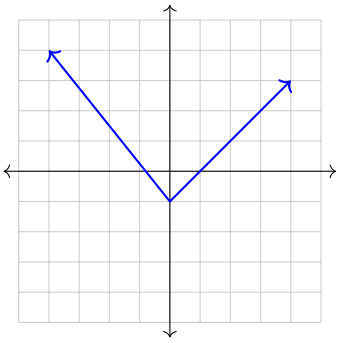
a)



Domain:

Range:

b)



Domain:

Range:

## Union and Intersection (Optional)

When working with multiple intervals, we sometimes want to express some combination of the intervals.

$A \cup B$: $A$ **union** $B$. This represents elements in $A$ **or** in $B$.

$A \cap B$: $A$ **intersection** $B$. This represents elements in $A$ **and** in $B$.

**Example 2.8.** Let $A$ be the interval $[3, 10)$ and $B$ be the interval $(-3, 5)$. Find the interval represented by

a) $A \cup B$

b) $A \cap B$

## Practice Exercises

1. Determine the domain and range of the relation.

   a) $(3,2),(2,3),(3,3)$

   Domain: _____

   Range: _____

   b) $(1,4),(4,1),(-1,4)$

   Domain: _____

   Range: _____

   c) $(-5,2),(0,2),(-5,3)$

   Domain: _____

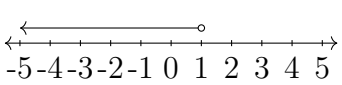
   Range: _____

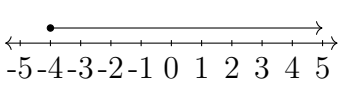
   d) $(5,-2),(3,2),(6,2)$

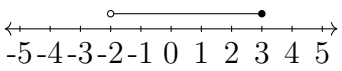
   Domain: _____

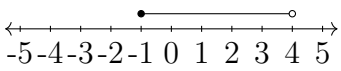
   Range: _____

2. Write the inequality represented by each number line.

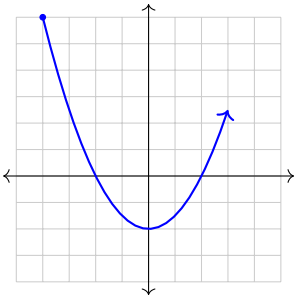
   a)
   
   

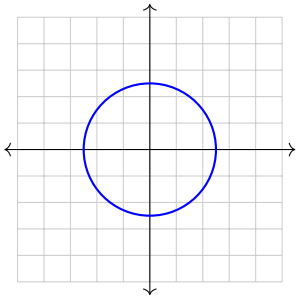
   b)
   
   

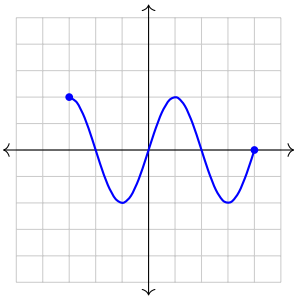
   c)
   
   

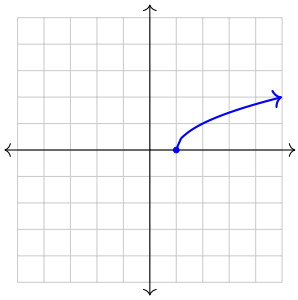
   d)
   
   

3. Find the domain and range of each graph. Express using **inequality notation**.

   a)
   

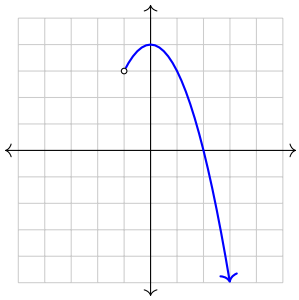
   b)
   

   c)
   

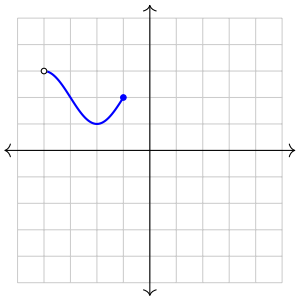
   d)
   

4. Find the domain and range of each graph. Express using **interval notation**.

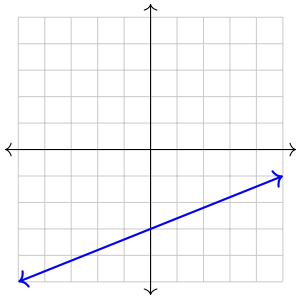
a)



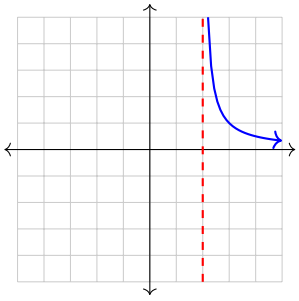
b)



c)

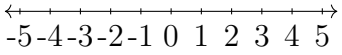


d)

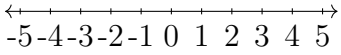


5. Plot the values on the number line indicated by each set notation.

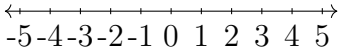
a) $\{x \mid -2 < x \leq 3\}$



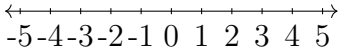

b) $\{x \in \mathbb{Z} \mid x > -3\}$




c) $\{x \mid x \neq 2\}$

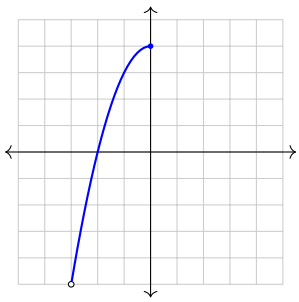



d) $\{x \in \mathbb{N} \mid x \geq 2\}$

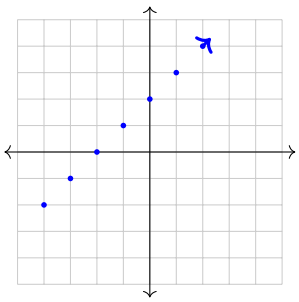



6. Find the domain and range of each graph. Express using **set-builder notation**.

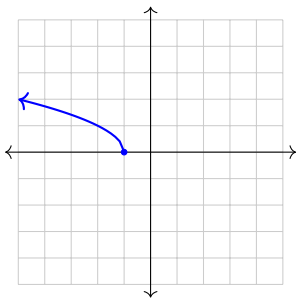
a)



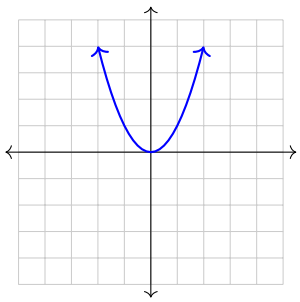
b)



c)

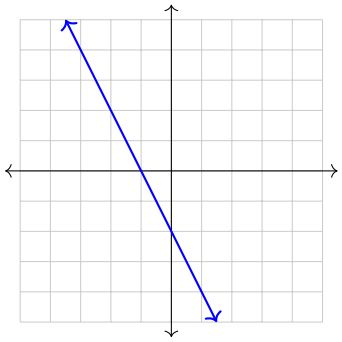


d)



7. Determine the domain and range. Also, determine if the relation is a function.

a)

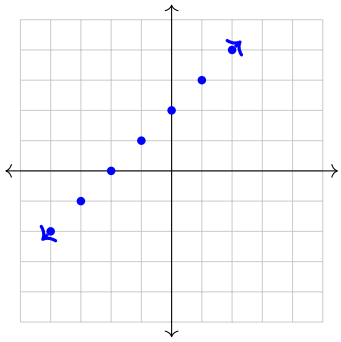


Domain: _____

Range: _____

Function (Y/N): _____

b)

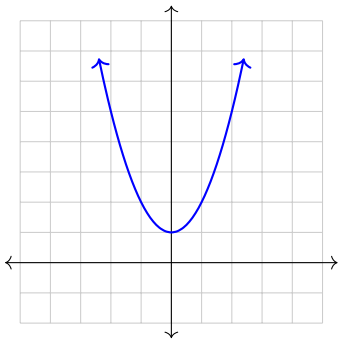


Domain: _____

Range: _____

Function (Y/N): _____

c)

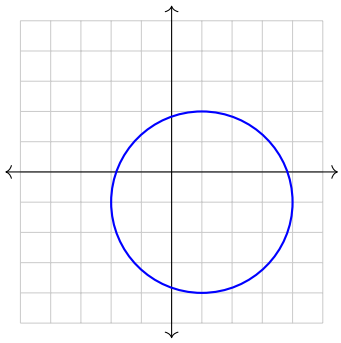


Domain: _____

Range: _____

Function (Y/N): _____

d)

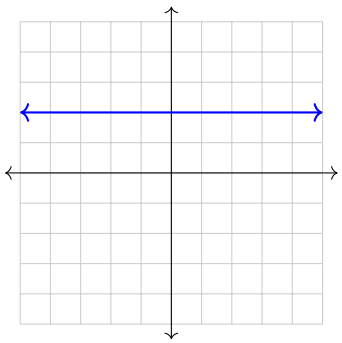


Domain: _____

Range: _____

Function (Y/N): _____

e)

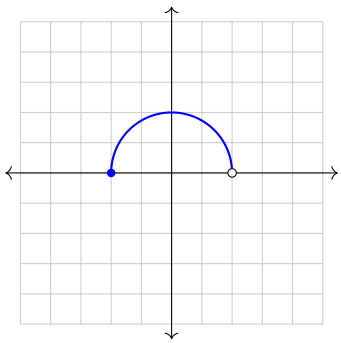


Domain: _____

Range: _____

Function (Y/N): _____

f)



Domain: _____

Range: _____

Function (Y/N): _____

8. Let $A$ be the interval $[2, 8)$ and $B$ be the interval $(-1, 5)$. Find the interval represented by

   a) $A \cup B$ b) $A \cap B$

9. Let $C$ be the interval $(-\infty, 4]$ and $D$ be the interval $[1, 7)$. Find the interval represented by

   a) $C \cup D$ b) $C \cap D$

10. Let $E$ be the interval $(-3, 0]$ and $F$ be the interval $[0, 4)$. Find the interval represented by

    a) $E \cup F$ b) $E \cap F$

11. Let $G$ be the interval $(5, \infty)$ and $H$ be the interval $[3, 8]$. Find the interval represented by

    a) $G \cup H$ b) $G \cap H$

12. Let $J$ be the interval $[-2, 3]$ and $K$ be the interval $(3, 6)$. Find the interval represented by

    a) $J \cup K$ b) $J \cap K$

# 3 Function Notation

If a relation is a function, we can write it using **function notation**.

The equation $y = 2x + 1$ represents a function relating $x$ (the **independent variable**) and $y$ (the **dependent variable**). In function notation, we write this as:

$$f(x) = 2x + 1$$

This is read as "$f$ of $x$" and means "the value of function $f$ at input $x$." For example, $f(3)$ represents the output when $x = 3$.

**Important:** The parentheses do *not* mean multiplication:

$$f(3) \neq f \times 3$$

They indicate the **input** to the function.

**Example 3.1.** Let $f(x) = 2x + 1$. Find the following

a) $f(-2)$

b) $f(3)$

**Example 3.2.** Let $g(x) = x^2 - 3$. Find the following

a) $g(-4)$

b) $g(0)$

**Example 3.3.** If $h(x) = 3 - 2x$. For what value(s) of $x$ will the following be true

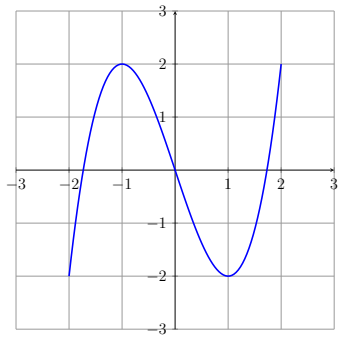
a) $h(x) = -17$

b) $h(x) = 13$

**Why use function notation?**

- It clearly distinguishes between different functions: $f(x) = 2x + 1$ vs $g(x) = -2x + 3$
- It emphasizes the input-output relationship
- It's the standard notation used in higher mathematics

**Example 3.4.** The graph of function $f$ is shown below. Use the graph to answer the questions.



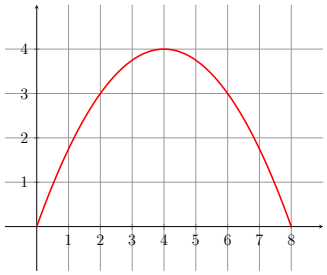

a) Find $f(1)$

b) Find $f(2)$

c) Find $x$ if $f(x) = -2$

d) Find $x$ if $f(x) = 2$

**Example 3.5.** The graph of function $g$ is shown below. Use the graph to answer the questions.




a) Find $g(2)$

b) Find $g(6)$

c) Find $x$ if $g(x) = 3$

d) Find $x$ if $g(x) = 0$

## Practice Exercises

1. If $f(x) = 5x - 7$, determine:

   a) $f(2)$ b) $f(-3)$ c) $f(0)$

2. If $f(x) = 8 - 2x$, determine:

   a) $f(4)$ b) $f(2t)$ c) $f(a+3)$

3. Complete the following.

   a) If $f(x) = 5x - 7$, determine the value of $x$ if $f(x) = 43$.

   b) If $g(x) = 6x + 6$, determine the value of $x$ if $g(x) = -24$.

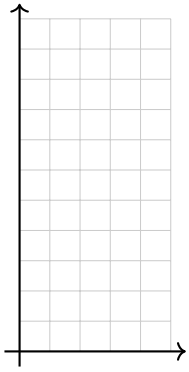
   c) If $g(x) = 56 - 3x$, determine the value of $x$ if $g(x) = 11$.

   d) If $h(x) = -3x + 1$, determine the value of $x$ if $h(x) = 22$.

4. The function $f(x) = 2x + 5$ has the domain $\{0, 1, 2, 3\}$

   (a) List the elements of the range.

   (b) Show the function using a graph.

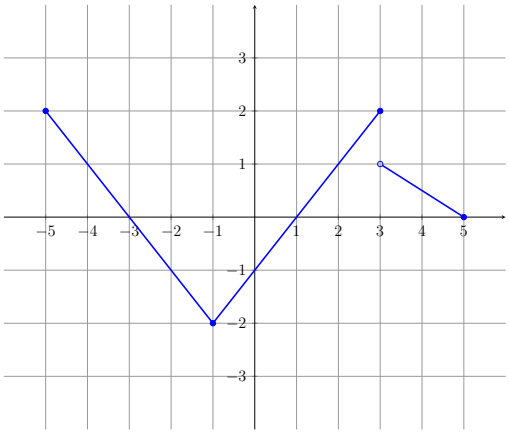


5. The domain of the function $h(x) = 2x - 6$ is $\{0, 5, 10, 15, 20\}$

   (a) List the elements of the range.

   (b) List the ordered pairs of the function.

6. The graph of function $f$ is shown below. Use the graph to answer the questions.



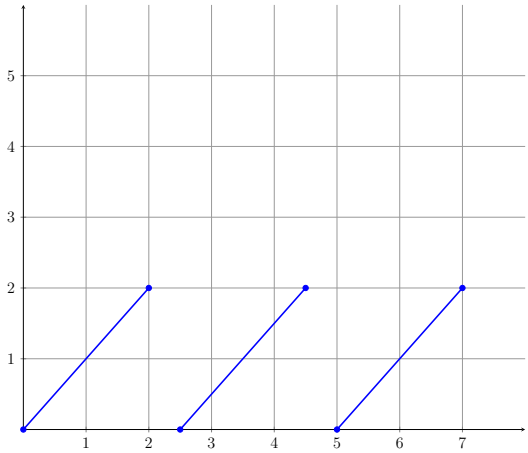

   a) Find $f(-3)$

   b) Find $f(0)$

   c) Find $x$ if $f(x) = 1$

   d) Find $x$ if $f(x) = -2$

7. The graph of function $h$ is shown below. Use the graph to answer the questions.




   a) Find $h(2)$

   b) Find $h(6)$

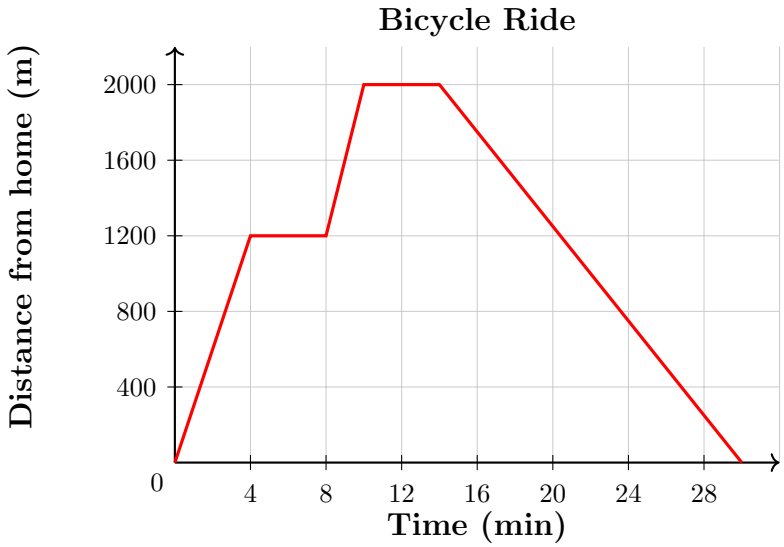
   c) Find $x$ if $h(x) = 2$

   d) Find $x$ if $h(x) = 0$

# 4 Interpreting Relations

Interpreting graphs in mathematics is a valuable skill.

**Example 4.1.** The graph shows the distance a bicyclist is from home.




a) How many minutes did the bike ride last?

b) When did the bicyclist start riding back home?

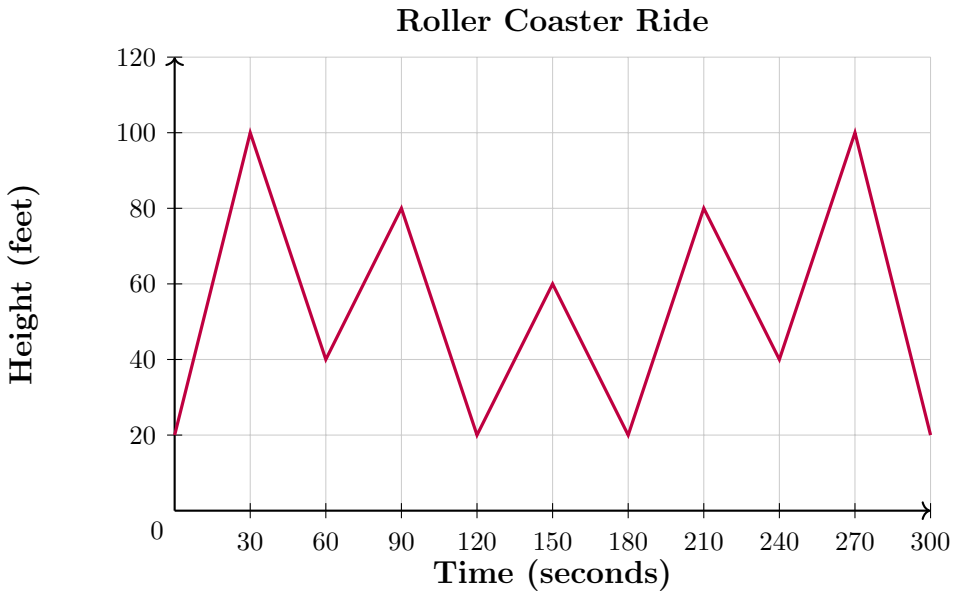
c) When did the bicyclist stop for a break?

d) What was average speed of the ride in the first four minutes? Last 16 minutes?

**Rate of Change**

The slope of a graph represent the rate of change.

**Example 4.2.** The graph shows the height of a roller coaster car throughout the ride.




a) How long did the entire roller coaster ride last?

b) What was the maximum height reached during the ride?

c) During which time interval was the roller coaster descending most rapidly?

d) How many times did the roller coaster reach a local maximum height?

**Instantaneous Rate of Change**

The steepness of a graph at any point represents the instantaneous rate of change.

**Example 4.3.** The graph shows the number of customers in a popular restaurant throughout the day.




a) During which hour was the restaurant most busy?

b) When did the lunch rush likely occur?

c) Estimate the rate of change in customers between 1pm and 3pm.

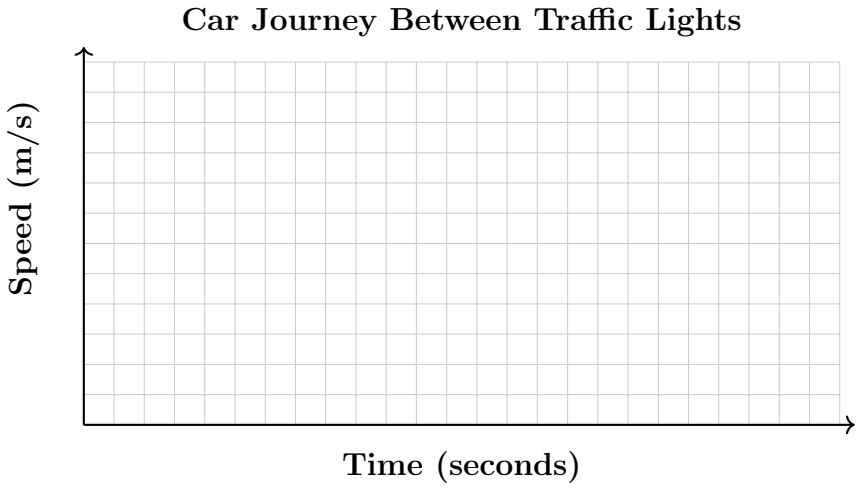
d) What might explain the dip in customers around 2pm?

**Interpreting Real-World Context**

Always consider what the graph represents in the real world when interpreting rates of change.

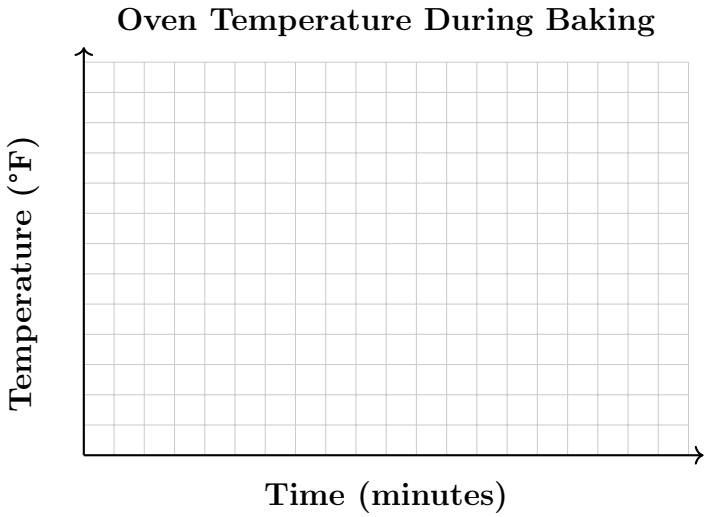
**Example 4.4.** A car starts from rest at a traffic light. It accelerates smoothly for 10 seconds, maintains a constant speed for 30 seconds, then decelerates smoothly to stop at the next red light over 15 seconds. After waiting 20 seconds, it accelerates again.

Sketch a graph of the car's speed as a function of time. Label each section clearly.



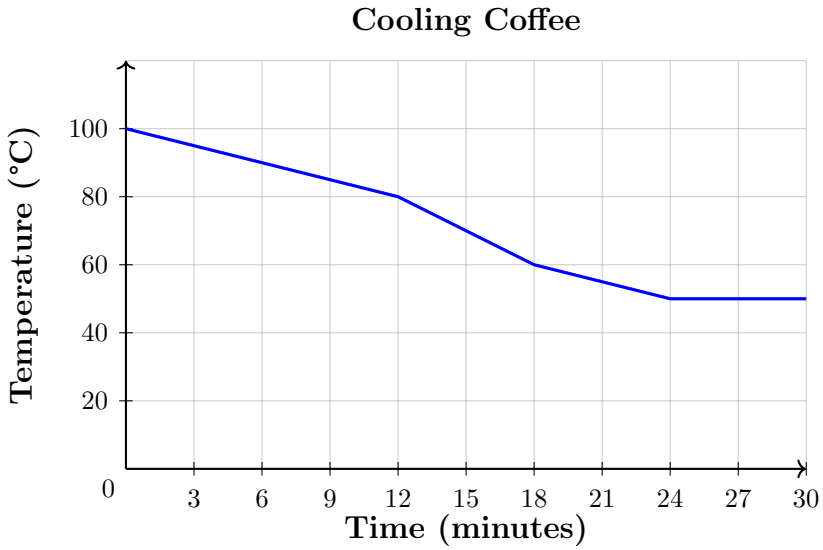

**Example 4.5.** Maya was baking cookies. She preheated the oven for 15 minutes until it reached 375°F, baked cookies for 12 minutes, then turned off the oven and let it cool down naturally. The oven temperature dropped slowly at first, then more rapidly after she opened the oven door to remove the cookies.

Sketch a graph of oven temperature as a function of time. Label each section of the graph and explain what it represents.



## Practice Exercises

1. The graph shows the temperature of a cup of coffee left on a desk over time.



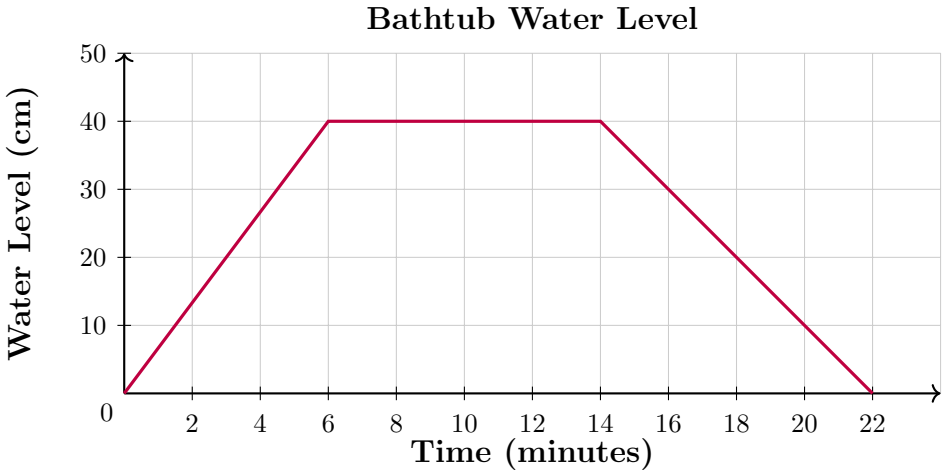

a) What was the initial temperature of the coffee?

b) During which time interval did the coffee cool most rapidly?

c) What appears to be the room temperature?

d) Estimate the average rate of cooling between 5 and 15 minutes.

2. Maria filled her bathtub and then took a bath. The graph shows the water level in the bathtub over time.



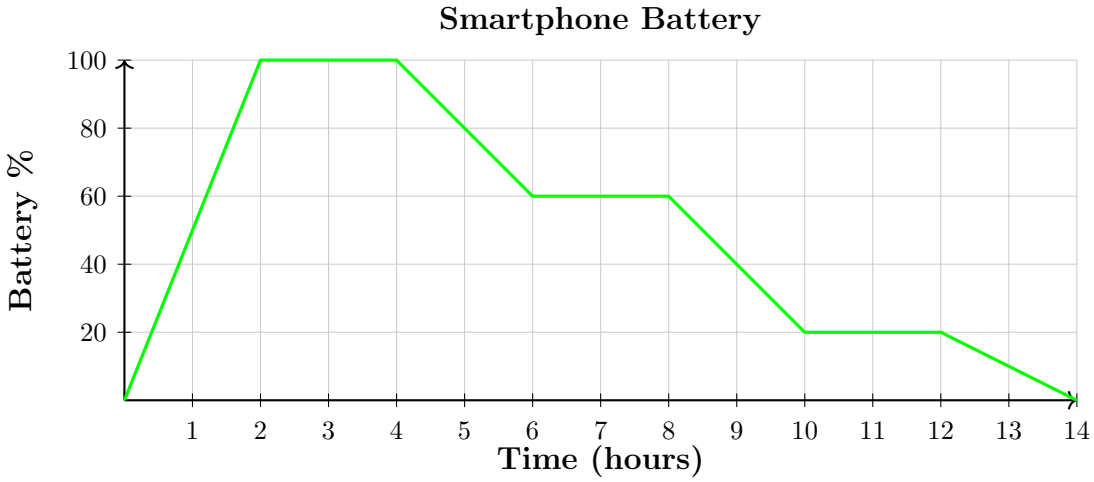

a) How long did it take to fill the tub?

b) How long did Maria spend in the bath?

c) When did she let water out of the tub?

d) At what rate was the tub draining? (cm/min)

3. A smartphone battery is charged from empty to full, then used throughout the day.




a) How long did it take to fully charge the phone?

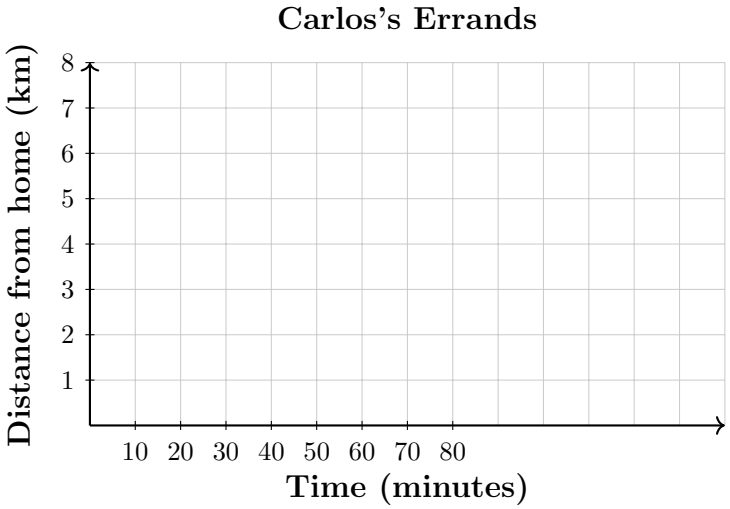
b) When was the phone being used most heavily?

c) How long did the phone stay at 100% charge?

d) Estimate how many hours of battery life the phone gets with normal use.

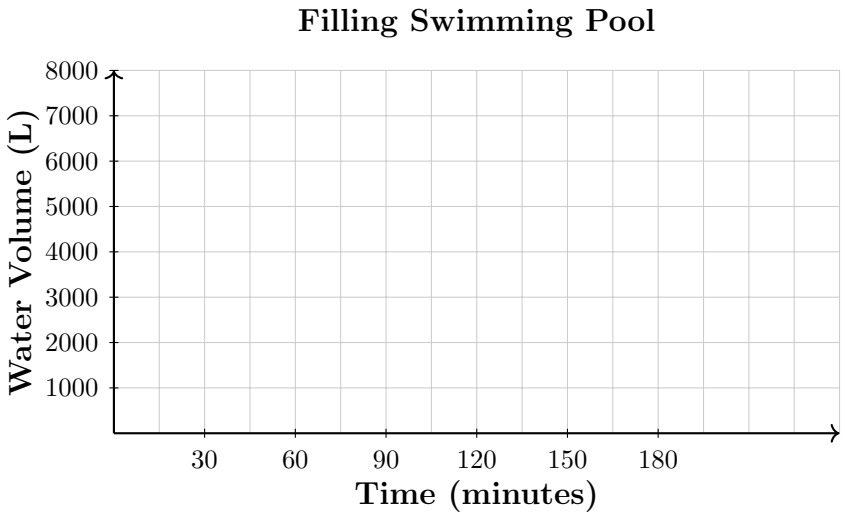
4. Carlos left home to run errands. Sketch a graph of his distance from home over time that matches this story:

   Carlos drove to the grocery store (15 minutes), shopped for 30 minutes, drove to the post office (10 minutes), spent 10 minutes there, then drove directly home (25 minutes).




5. A swimming pool is being filled with water. The rate of filling changes at different times. Sketch a graph of the water volume in the pool over time that shows:

   - Slow filling for the first 30 minutes
   - Fast filling for the next 45 minutes
   - No filling for 15 minutes (lunch break)
   - Medium filling for the final 60 minutes until full



# 5 Arithmetic Sequence and Series

A **sequence** is simply a list of numbers. In a sequence each number is called a **term** of the sequence. There is a first term, second term, third term, and so on. A sequence can be **finite**, in which it is possible to count the number of terms, or **infinite**, in which the terms continue forever.

| | |
|---|---|
| $1, 3, 6, 10$ | is a finite sequence. |
| $1, 3, 6, 10, \ldots$ | is an infinite sequence. |

## Notation

There are some common notation that we use when describing sequences, terms, and position.

$$a_1, a_2, a_3, \ldots, a_n$$

In the above, each $a$ is differentiated by a subscript. The subscript determines the position of the term, and hence needs to be a **natural number**.

In this course, we will use try to use $a_n$ as our notation to represent the $n^{th}$ element. It is also common to see $t_n$ and $u_n$ as other variable choices.

Sometimes a sequence is defined as a function of $n$.

**Example 5.1.** Write the first 4 terms of the sequence.

a) $a_n = \dfrac{n+1}{n}$ b) $a_n = 2n - 3$ c) $a_n = 2^n$

When the difference between successive terms of a sequence is always constant, the sequence is called an **arithmetic sequence.** For example, the sequence $3, 7, 11, 15, ...$ is arithmetic because adding 4 to any term produces the next term. The **common difference**, $d$, of this sequence is 4.

**The $n^{th}$ Term of an Arithmetic Sequence**

We define the general term of an arithmetic sequence as

$$a_n = a + (n-1)d$$

where

$a_n$ : the nth term
$a$ : the first term
$n$ : the position or number of terms
$d$ : the common difference

**Example 5.2.** For each arithmetic sequence, determine the first term, $a$, and the common difference, $d$.

a) $3, 5, 7, 9, \ldots$

b) $11, 8, 5, 2, \ldots$

**Example 5.3.** Find the $12^{th}$ term of the arithmetic sequence $2, 5, 8, \ldots$

**Example 5.4.** Which term in the arithmetic sequence $4, 7, 10, \ldots$ has a value of 439?

The general term equation for an arithmetic sequence, is actually a linear function.

**Example 5.5.** Given the arithmetic sequence

$$4, -3, -10, -17, ...$$

a) Find and simplify the equation of the general term, $a_n$

b) Compare it to the slope-intercept form $y = mx + b$ and state the slope and the value of the $y$-intercept.

## Arithmetic Series

A **series** is taking the sum of an indicated number of terms in a sequence.

$$\begin{aligned} 3, 7, 11, 15 : &\quad \text{a sequence} \\ 3 + 7 + 11 + 15 : &\quad \text{a series} \end{aligned}$$

We will strictly be working with finite series, but there exists infinite series as well.

**Sum of an Arithmetic Series**

The sum of the first $n$ terms of an arithmetic series can be calculated by:

$$S_n = \frac{n}{2}(a + l) \tag{1}$$

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right] \tag{2}$$

where

$S_n$ : the sum of the first $n$ term
$a$ : the first term
$l$ : the last term
$n$ : the position or number of terms
$d$ : the common difference

**Example 5.6.** Find the sum of the positive integers from 1 to 50 inclusive.

**Example 5.7.** Find the sum of the first 25 terms of the series $11 + 15 + 19 + \ldots$

**Example 5.8.** Find the sum of the series $100 + 97 + 94 + \cdots + 7$

**Example 5.9.** A theatre has 25 seats in the first row, 27 seats in the second, 29 in the third row, and so on. If there are 30 rows of seats, how many seats are there?

## Practice Exercises

1. For the following arithmetic sequence, determine the common difference and find the next three terms of the sequence.

   a) $8, 14, 20, \ldots$

   b) $-5, 7, 19, \ldots$

   c) $70, 53, 36, 19, \ldots$

   d) $\frac{2}{3}, \frac{1}{15}, -\frac{8}{15}, \ldots$

2. Calculate the first four terms of the arithmetic sequence with the given term and the common difference.

   a) $a_1 = 5, \quad d = 6$

   b) $a_3 = 15, \quad d = -2$

   c) $a_5 = 20, \quad d = -1$

3. Consider the arithmetic sequence $12, 5, -2, -9, \ldots$

   a) Determine the general term.

   b) Determine the $19^{th}$ term.

4. Determine the number of terms in each arithmetic sequence.

   a) $4, 7, 10, \ldots, 49$

   b) $-52, -56, -60, \ldots, -148$

5. A contractor charges \$68 for the first hour of work. This includes a rate of one hour of work and a fixed travel fee. For a two-hour job the contractor charges \$110 and for a three-hour job the contractor charges \$152.

   (a) State the first three terms of the sequence using the above fees.

   (b) State the rate for any additional hour of work.

   (c) Determine the amount of the travel fee.

   (d) Determine the cost for a 6-hour job.

6. Find the sum of each arithmetic series.

a) $2 + 3 + 4 + \cdots + a_{30}$

b) $(-8) + (-4) + 0 + \cdots + a_{27}$

c) $2.5 + 2.7 + 2.9 + \cdots + a_{16}$

d) $\frac{5}{2} + \frac{11}{6} + \frac{7}{6} + \frac{1}{2} + \cdots + a_{12}$

7. Find the sum of each arithmetic series given the first and last term

a) $a_1 = 8, \quad a_{15} = 120$

b) $a_1 = -11, \quad a_{23} = -253$

8. Find the sum for each arithmetic series.

a) $11 + 23 + 35 + 47 + \cdots + 179$

b) $29 + 21 + 13 + 5 + \cdots + (-27)$

9. Fred starts work at a salary of \$72,000 annually. He receives annual increases of \$1950. He has worked at the company for 13 years.

   (a) Calculate his salary in the tenth year.

   (b) How much has he earned in total over twelve years?

10. Find the sum of all multiples of 7 between 50 and 500.

# 6 Reasoning and Analyzing

Each statement is either **always true, sometimes true,** or **never true**. When you decide on your answer, you must provide a **justification**.

If you choose **sometimes true**, you must show one scenario when it is true, and another, when it is false.

For **always true** and **never true**, you must provide a brief explanation why. Sometimes this can simply be a definition, or sometimes it may require some more mathematical reasoning.

**Example 6.1.** For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

a) A function is a relation.

b) A relation is a function.

c) If $f(3) = 4$, then $(3, 4)$ is a point on the function.

d) For $a < b < c$, the two intervals $[a, b]$ and $[b, c]$ do not overlap.

e) The sequence $4, 7, 10, 13, \ldots$ is an arithmetic sequence.

f) The sum of an arithmetic sequence, $S_n$, is greater than than the value of the first term, $a_1$.

**Example 6.2.** A student was given the function $f(x) = x^2 - 3$ and $g(x) = 5 - 3x$. They were then asked to evaluate $f(-2) + g(-2)$. Which step did they first make an error? Explain and correct.

$$
\begin{aligned}
& f(-2) + g(-2) \\
=& \Big[(-2)^2 - 3\Big] + \Big[5 - 3(-2)\Big] && \textbf{Step 1} \\
=& \Big[-4 - 3\Big] + \Big[5 + 6\Big] && \textbf{Step 2} \\
=& -7 + 11 && \textbf{Step 3} \\
=& 4 && \textbf{Step 4}
\end{aligned}
$$

**Example 6.3.** In each of the following, one item doesn't belong...

a) Which sequence doesn't belong? Explain.

i. $1, 1, 1, 1$ ii. $2, 6, 10, 14$ iii. $2, 4, 8, 16$ iv. $-1, -10, -19, -28$

b) If $f(x) = x - 5$, which expression don't belong? Explain.

i. $f(0)$ ii. $f(-2)$ iii. $f(3)$ iv. $f(5)$

## Practice Exercises

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) A relation with exactly 3 elements in the domain and exactly 1 element in the range is not a function.

   (b) Let $a_1, a_2, a_3$ represent an arithmetic sequence, the sum of the three terms does not equal 0.

   (c) If $f(x) = 2x - 5$, then $f(6) = 7$.

   (d) $x \in [6, 7)$ is equivalent to $6 < x \leq 7$.

   (e) $\{x \in \mathbb{Z} \mid 3 < x \leq 10\}$ has exactly 7 elements in its set.

   (f) For distinct $a, b, c, d$, $[a, b] \cap [c, d] \neq [a, b]$.

2. A student was given $i(x) = 7 - 2x$ and was asked to solve for $x$ given that $i(x) = 10$. Which step did the student first make an error? Explain and correct.

$$\begin{aligned} i(x) &= 7 - 2x \\ i(10) &= 7 - 2(10) && \textbf{Step 1} \\ &= 7 - 20 && \textbf{Step 2} \\ &= -13 && \textbf{Step 3} \end{aligned}$$

3. In each of the following, one item doesn't belong...

a) Which domain does not belong? Explain.

i. $2 < x < 5$
ii. $x \geq 4$
iii. $x \leq -5$
iv. $\{x \in \mathbb{N} \mid 2 < x < 5\}$

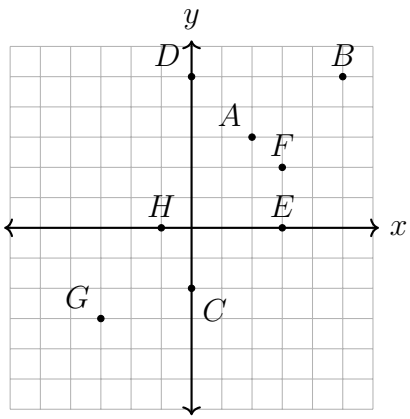
b) If $f(x) = x(x + 6)(x - 7)$, which value doesn't belong? Explain.

i. $f(-6)$
ii. $f(0)$
iii. $f(7)$
iv. $f(6)$

# 7 Answer Key

## Section 1 - Exercises

1.




2. $A(5,5)$ $B(4,0)$ $C(0,5)$ $D(0,-2)$ $E(-2,0)$ $F(-5,3)$ $G(6,-5)$ $H(-3,-6)$

3. A

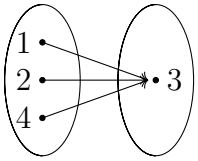
4. C

5. D

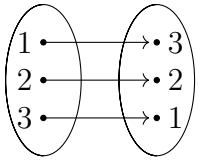
6. a) $(-1,2)$
$(3,-4)$

b) $(5,-3)$
$(5,2)$

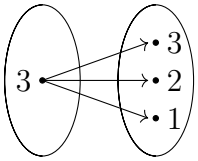
7. a)



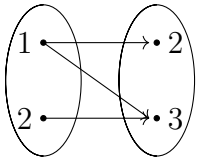

b)




c)




d)

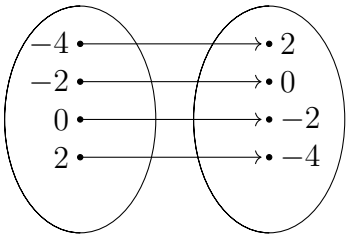


8. B 9. D 10. A 11. B

12. a) No b) No c) Yes d) Yes

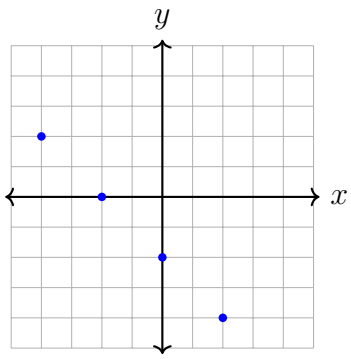
13. a) Yes b) Yes c) No d) No

14. a)

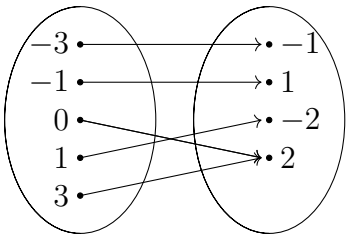



b) $(4,-2),(-2,0),(0,-2),(2,-4)$

c)




15. a)




b) $(-3,-1),(-1,1),(0,2),(1,-2),(3,2)$

c)

| $x$ | -3 | -1 | 0 | 1 | 3 |
|---|---|---|---|---|---|
| $y$ | -1 | 1 | 2 | -2 | 2 |

## Section 2 - Exercises

1. a) Domain: $\{3, 2\}$
Range: $\{2, 3\}$

   b) Domain: $\{1, 4, -1\}$
Range: $\{4, 1\}$

   c) Domain: $\{-5, 0\}$
Range: $\{2, 3\}$

   d) Domain: $\{5, 3, 6\}$
Range: $\{-2, 2\}$

2. a) $x < 1$

   b) $x \geq -4$

   c) $-2 < x \leq 3$

   d) $-1 \leq x < 4$

3. a) Domain: $x \geq 4$
Range: $y \geq -2$

   b) Domain: $-2.5 \leq x \leq 2.5$
Range: $-2.5 \leq y \leq 2.5$

   c) Domain: $-3 \leq x \leq 4$
Range: $-2 \leq y \leq 2$

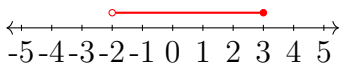
   d) Domain: $x \geq 1$
Range: $y \geq 0$

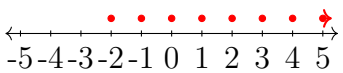
4. a) Domain: $(-1, \infty)$
Range: $(-\infty, 4)$

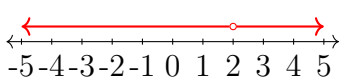
   b) Domain: $(-4, -1]$
Range: $[1, 3)$

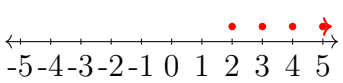
   c) Domain: $(-\infty, \infty)$
Range: $(-\infty, \infty)$

   d) Domain: $(2, \infty)$
Range: $(0, \infty)$

5. a)



   b)



   c)



   d)



6. a) Domain: $\{x \mid -3 < x \leq 0\}$
Range: $\{y \mid -5 < y \leq 4\}$

   b) Domain: $\{x \in \mathbb{Z} \mid x \geq -4\}$
Range: $\{y \in \mathbb{Z} \mid y \geq 2\}$

   c) Domain: $\{x \mid x \leq -1\}$
Range: $\{y \mid y \geq 0\}$

   d) Domain: $\{x \in \mathbb{R}\}$
Range: $\{y \mid y \geq 0\}$

7. a) Domain: $x \in \mathbb{R}$
Range: $y \in \mathbb{R}$
Function: Yes

   b) Domain: $x \in \mathbb{Z}$
Range: $y \in \mathbb{Z}$
Function: Yes

   c) Domain: $x \in \mathbb{R}$
Range: $y \geq 1$
Function: Yes

   d) Domain: $-2 \leq x \leq 4$
Range: $-4 \leq y \leq 2$
Function: No

   e) Domain: $x \in \mathbb{R}$
Range: $\{2\}$
Function: Yes

   f) Domain: $-2 \leq x < 2$
Range: $0 \leq y \leq 2$
Function: Yes

8. a) $(-1, 8)$

   b) $[2, 5)$

9. a) $(-\infty, 7)$

   b) $[1, 4]$

10. a) $(-3, 4)$

    b) $[0, 4)$

11. a) $[3, \infty)$

    b) $(5, 8]$

12. a) $[-2, 6)$

    b) $\varnothing$

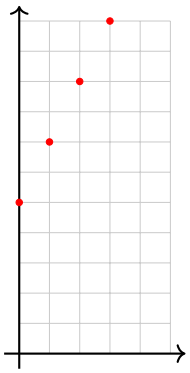
## Section 3 - Exercises

1. a) 3 b) $-22$ c) $-7$

2. a) 0 b) $8 - 4t$ c) $-2a + 2$

3. a) $x = 10$ b) $x = -5$ c) $x = -15$ d) $x = -7$

4. a) $\{5, 7, 9, 11\}$ b)



5. a) $\{-6, 4, 14, 24, 34\}$ b) $(0, -6), (5, 4), (10, 14), (15, 24), (20, 34)$

6. a) 0 b) $-1$ c) $x = 2, -4$ d) $x = -1$

7. a) 2 b) 1 c) $x = 2, 4.5, 7$ d) $x = 0, 2.5, 5$

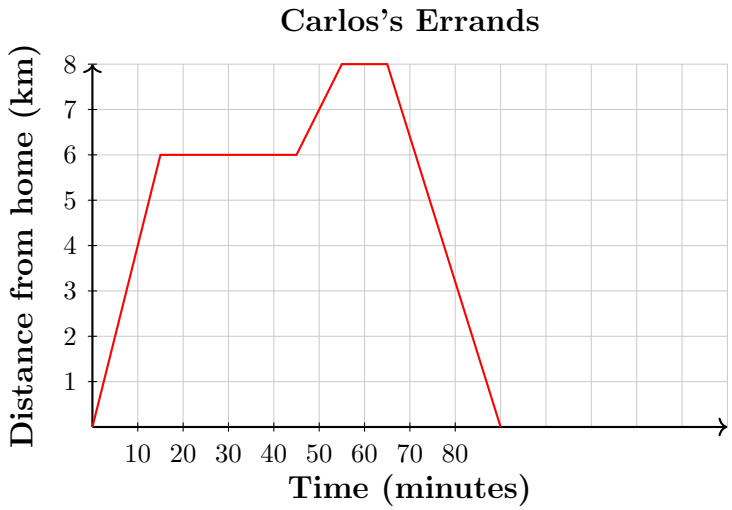
## Section 4 - Exercises

1. a) $100°C$ b) About 12.5 to 17.5 mins
   c) $50°$ d) $-2°C/\text{min}$

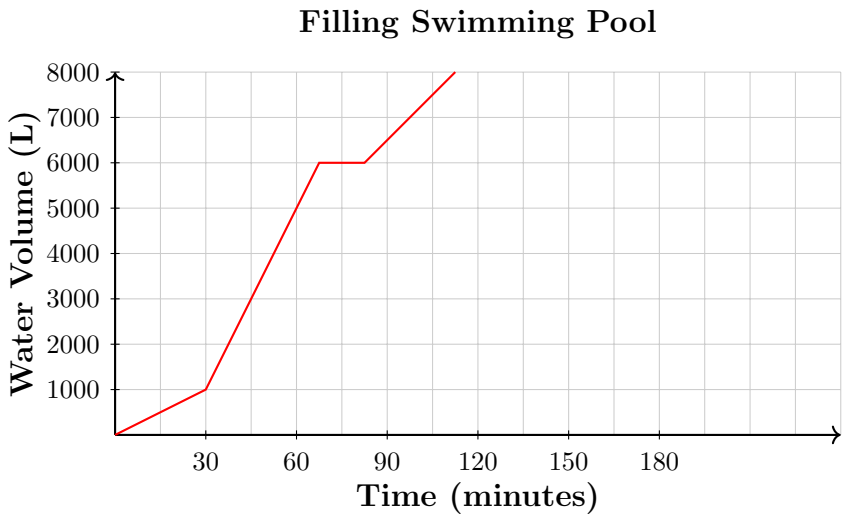
2. a) 5 mins b) About 7 mins
   c) Just before the 12 minute mark d) $\approx -6.67$ cm/min

3. a) 2 hrs b) Between 4-6 hrs and 8-10 hrs
   c) 2 hrs d) $\approx 12$ hrs

4. Answers may vary a bit




5. Answer may vary a bit




## Section 5 - Exercises

1. a) $d = 6$, $26, 32, 38$
   b) $d = 12$, $31, 43, 55$
   c) $d = -17$, $2, -15, -32$
   d) $d = -\frac{3}{5}$, $-\frac{17}{15}, -\frac{26}{15}, -\frac{7}{3}$

2. a) $5, 11, 17, 23$ b) $19, 17, 15, 13$ c) $24, 23, 22, 21$

3. a) $a_n = 19 - 7n$ b) $a_{19} = -114$

4. a) $n = 16$ b) $n = 25$

5. a) $68, 110, 152$ b) \$42/hour
   c) \$26 d) \$278

6. a) $S_{30} = 495$ b) $S_{27} = 1188$
   c) $S_{16} = 64$ d) $S_{12} = -14$

7. a) $S_{15} = 960$ b) $S_{23} = -3036$

8. a) $S_{15} = 1425$ b) $S_8 = 8$

9. a) \$89 550 b) \$992 700

10. $S_{64} = 17696$

## Section 6 - Exercises

1. a) Never true. It is always a function since all the domain elements can only go to exactly one range element. In other words, no single domain element goes to more than one range element.

   b) Sometimes true. Depends on the terms and difference, for example $-4, 0, 4$ is a sequence that sums to 0, whereas $2, 3, 4$ does now.

   c) Always true. We see that $f(6) = 2(6) - 5 = 7$.

   d) Never true. The square end bracket means include and open end bracket means not include. So the correct inequality is $6 \leq x < 7$.

   e) Always true. We are working with integers only, not including 3, we have $\{4, 5, 6, 7, 8, 9, 10\}$. Exactly 7 elements.

   f) Sometimes true. If $a, b$ are within $c, d$, then we can have it true. For example, if $a = 2, b = 3, c = 0, d = 4$. then $[2, 3] \cap [0, 4] = [2, 3]$. An example of not could be where $a, b$ is outside $c, d$. $a = 1, b = 3$ and $c = 2, d = 4$.

2. The mistake happened in step 1. The student inserted $x = 10$ as the input, whereas 10 is the output. They should be solving for $x$ in $7 - 2x = 10$.

3. a) iii. doesn't belong because it includes negative numbers
   iv. doesn't belong because it has a finite number of elements

   b) i. doesn't belong because it's input is negative
   iv. doesn't belong because it does not equal 0
   ii. doesn't belong because it is the only 0 input